AdvPrint Pro

Advanced Printer — silent printing for every web page, PDF, application and receipt printer

by Bit Dream IT — Innovative Healthcare Solutions

Version 2.7.0 — Enterprise Edition

Full Setup Guide — A to Z

100% local · no cloud · no tracking · MIT licensed

Windows 7/8/10/11 · Windows Server · Debian/Ubuntu/Mint · macOS

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1. What AdvPrint Pro is — and how one print job travels

AdvPrint Pro (Advanced Printer) by **Bit Dream IT — Innovative Healthcare Solutions** prints anything — PDFs, web pages, `window.print()` calls, receipts (raw/ESC-POS), screenshots, selections — **silently**, with no print dialog, on any printer connected to the hub PC, including printers shared from other PCs. Everything stays on your machines: the hub listens on `127.0.0.1:9235` only (until you deliberately enable LAN sharing), no cloud, no telemetry.

> Product identity: **AdvPrint Pro** — *AdvPrint (Advanced Printer)* — **by Bit Dream IT — Innovative Healthcare Solutions**. The hub reports `app: "AdvPrint Pro"` and the full brand string in every /health answer.

```
Chrome (any web page)
   │  extension: auto-print rules / right-click / Alt+P / window.advprint API
   ▼
AdvPrint Pro HUB  (http://127.0.0.1:9235, token-protected)
   │  queue → retries → holds → printer status checks
   ▼
Silent print engines:  Windows: SumatraPDF (PDF) + Chrome headless (pages) + raw ESC/POS
                       Linux/macOS: CUPS lp + Chrome headless
   ▼
Your printer — or ~/.advprint/files if the destination is "save as PDF"
```

- **Hub** = tiny local web server (`advprint.py`) that owns the queue, the history and the printers.
- **Extension** = the browser side: auto-print rules, popup controls, the Options page (tab-wise), page API.
- **Dashboard** = the hub's browser UI at `/dashboard` : status, queue, history, audit, debug, LAN, live test, settings.
- **Tray icon** (Windows) = one click to the dashboard, pause/resume, test page, auto-run toggle.
- **Desktop tool** = `AdvPrint Tool` (Start Menu) — numbered menu for start/stop/service/live test/updates plus **[13] Diagnose & fix** (repairs “hub offline” automatically).

2. Install in 3 minutes

2.1 Windows — the one-time installer (recommended)

1. Download `AdvPrint-Pro-v2.7.0-FULL.zip` and unzip it.
2. Run `AdvPrint-Pro-Setup-2.7.0.exe` . No admin needed, no UAC popup, no Python needed — the setup carries a **complete Python 3.11.9 runtime** (this is exactly what the broken 2.5.3 download missed: if an installer is smaller than ~9 MB it is incomplete — re-download).
3. Setup registers the **Startup app "AdvPrint Pro"** (Task Manager → Startup apps), creates Start Menu shortcuts, and starts the hub. A tray icon `A` appears next to the clock.
4. Optional ticks: **Run as Windows service** (headless, services.msc) and **Install browser extension files**.

> Windows says *"cannot access / Access is denied"* on a downloaded setup? That is Mark-of-the-Web on a damaged/incomplete download. The 2.6.x setup is complete (~9.6 MB) and the portable `install\install-windows.bat` unblocks every file automatically. If your AV quarantines the hub, whitelist the folder `%LOCALAPPDATA%\AdvPrintPro` — the hub is a plain stdlib Python script, signed-off and open source (MIT).

2.2 Windows — portable zip (no installer)

1. Unzip. Double-click `tool\windows\portable\install\install-windows.bat` . It unblocks files, frees port 9235, uses the bundled runtime, registers auto-run, health-checks the hub.
2. Undo any time: `UNINSTALL-WINDOWS.bat` .

2.3 Linux / Debian / Ubuntu / Mint — the .deb

```
sudo apt install ./advprint-pro_2.7.0_all.deb
advprint-start           # start now
advprint token           # show the current token
advprint livetest        # full A-to-Z self-test
```

Auto-run at login: run `/opt/advprint/install/install-unix.sh` (installs a **systemd --user** service — no root needed; falls back to XDG autostart). Headless server mode: `sudo /opt/advprint/install/install-unix.sh --service` .

2.4 macOS

Run the portable hub: `python3 tool/unix/hub/advprint.py serve`, then `install-unix.sh` (LaunchAgent auto-run). CUPS printers are used through `lp`; page printing uses Chrome/Edge if installed.

2.5 First print in 60 seconds

1. Hub running (tray icon green / `advprint status`).
2. Load the extension: `chrome://extensions` → Developer mode → **Load unpacked** → the `extension` folder (or the one installed to `%LOCALAPPDATA%\AdvPrintPro\extension`).
3. Open the extension Options → Settings → **Detect token** (one click) → **Connect** shows `CONNECTED`.
4. Click **Send test page**. Paper comes out. Done.

3. The token — and the one-click fix

Every request to the hub carries a token. Since v2.0 a **fresh install generates a private random token** and stores it in `%USERPROFILE%\.advprint\config.json` (Linux/macOS `~/.advprint/config.json`). Upgrades of pre-2.0 installs keep working with the old default `PrintHub-2024` until you rotate it — the hub warns you in the dashboard until then. Pre-2.6 installs with a `~/.printhub` folder are **migrated automatically** to `~/.advprint` on first start (old folder kept as backup).

3.1 The one-click fix (v2.6)

Open a stale bookmark like `/dashboard?token=PrintHub-2024` and you get the friendly error page — with a green **"Fix it now — open the dashboard with the current token"** button. One click and you are in. It works because the hub hands the current token only to programs **on the hub PC itself** (loopback-only `GET /token/current`); other PCs on the LAN still see the manual steps only. The same button exists in the extension Options → Settings → **Detect token**.

3.2 Every way to get the current token

Way	How
Tray icon (easiest)	Right-click the **AdvPrint Pro** tray icon → **Open dashboard** — always opens with the right token.
Dashboard error page	Green **Fix it now** button (hub PC only) — or **Copy current token**.
Extension Options	Settings → **Detect token** button fills and saves it automatically.
Command line	`python advprint.py token` (portable) / `advprint token` (deb).
Config file	`"token"` value in `~/.advprint/config.json`.

3.3 Rotating the token

```
python advprint.py token-rotate        # CLI
POST /token/rotate (header auth)       # API
Dashboard → "Rotate token" button
```

After rotating: press **Detect token** in the extension (or paste the new token), and update old dashboard bookmarks. Rotating **invalidates the old token immediately** — including `PrintHub-2024` on upgraded installs.

4. The browser extension — every Options tab explained

Open the extension icon → ⚙ (or right-click the toolbar icon → Options). The Options page is **tab-wise**: `Queue` `History` `Settings` `Debug` `LAN / PCs` `About`. The extension opens on Queue when something is waiting, otherwise on Settings. Every tab auto-refreshes live.

4.1 Queue tab — live FIFO

- Hub state dot: running / PAUSED / offline, pending count, jobs today, hub version.
- Every waiting job row: id, title, printer, kind, status chip + buttons: **Cancel**, **Shift to PC** (print this job on another PC's hub over the LAN), **Print now** (bypasses pause and printer-status hold — the remedy when jobs sit "stuck").
- **Pause / Resume** queue, **Clear** queue, **Refresh**. Auto-refresh every 3 s.

4.2 History tab — last 100 jobs

- Every job with time, title, printer, kind and a colored status chip — **queued** gray, **printed** green, **failed/dead/canceled** red, **held** orange (with the exact held reason, e.g. "printer offline").
- Row actions: **Retry** (failed jobs — optionally on another printer), **Reprint** (printed jobs — reason goes to the audit log), **Cancel/Print now** (waiting jobs).
- History auto-refreshes every 8 s — a stuck "queued" row turns "printed" in front of you.
- “Clear” wipes only the browser-side copy; the hub history stays.

4.3 Settings tab — all switches

- **Power switches**: master **AdvPrint enabled** (OFF = nothing prints, gray OFF badge), **Queue mode** (ON = FIFO order + pause; OFF = instant print with auto-queue fallback), **Debug mode**, **Retry when the hub comes back**, UI language (English / বাংলা / auto).
- **Connection**: Hub URL (`http://127.0.0.1:9235`), Token + **Connect** + **Detect token**; buttons to open the dashboard and the audit log.
- **Printing**: default printer (or "File (save as PDF)"), plus **default print options** sent with every job: copies, pages (odd/even/custom), quality (draft…1200 dpi), per-sheet n-up, duplex, layout, scale, margins, paper size — with live capability hints from the selected printer.
- **Auto-print & routing rules**: URL pattern (glob or regex), page-title filter, per-rule printer, action (auto/PDF/page), per-site ON/OFF, auto-print-on-open, delete. First match wins.
- **Behavior**: print every PDF opened in the browser (simple mode), intercept `window.print()` (also Ctrl+P), close tab after auto-print, failure fallback (new tab vs silent), right-click menu, toolbar badge.
- **Hub settings (advanced)** — v2.6: the hub's own switches, editable live (see §9 for the full list): queue & retries, security/lockout, reprint watermark & reason, audit, thermal text engine + ESC/POS codepage, save-as-PDF folder, keep-files, SumatraPDF path.

4.4 Debug tab — WHEN / WHY / WHERE

- One card: hub version, debug mode on/off, worker alive, queue paused, printers ready, instant auto-queue state.
- **Problems found by the hub** table with the exact fix for each finding.
- **Browser error reports**: everything this browser hit (network, token, page JS) while Debug mode was on.
- Buttons: Refresh · Send test report · Copy report (everything for a bug report) · Open /debug JSON · **Run full self-test (A-to-Z)** — runs the hub's live test right here and prints OK/!! lines (v2.6).

4.5 LAN / PCs tab — share & shift

- Shows this PC's LAN URL (`http://192.168.x.x:9235`) when sharing is ON, with copy button.
- **Enable/Disable LAN sharing** (writes `share_lan` ; restart the hub to rebind + open TCP 9235 in the firewall).
- **Other PCs (peers)**: online/offline + version of each configured peer hub, one-click **Shift all waiting jobs**. Or type a LAN URL and shift everything ad hoc.

4.6 About tab

Version + full brand line, update check against GitHub releases, links to the user guide served by the hub (`/docs`), the developer API (`window.advprint`) and the test tools.

4.7 The popup (toolbar icon)

Master switch, queue mode switch, live queue state, pause/resume, last jobs, test page, screenshot-to-printer, save-page-as-PDF, offline banner with retry counter, token warnings with the exact remedy. Hotkey `Alt+P` prints the current tab. Right-click menu prints page / selection / linked PDF / screenshot.

> **v2.7.0 — popup LAN sharing switches:** the toolbar popup now has a **LAN sharing** section with **Share Receive Mode** (this PC accepts jobs from other PCs = hub `share_lan` , saved straight into the hub config) and **Share Send Mode** (this browser may shift jobs to peer PCs — OFF locks every send-to-PC button here). Both turn gray while the hub is offline and exist in English + Bangla. Nothing was removed anywhere.

5. The hub dashboard — every card and button

Open `http://127.0.0.1:9235/dashboard?token=…` (the tray icon does this with one click). All cards refresh live (5–15 s).

Area	What it shows / does
Status cards	Success rate (7 days) · queue size · jobs today · printers ready · hub version.
Action bar	Pause/Clear queue · Docs (user guide) · History CSV · Audit CSV · Diagnostics JSON · Debug · Health JSON · **Rotate token**.
Token warning	Appears while the old default `PrintHub-2024` is active, with a one-click **Rotate now**.
LAN sharing & other PCs	Enable/disable + this PC's LAN URL + printer-status-hold switch + peer PCs with “Shift all waiting jobs”.
Debug card	Debug mode toggle, worker/queue state, problems with fixes, client error reports, copy report.
Printers	Every printer + live status (ready / offline / paper-out / error).
Queue (waiting + printing)	Cancel · Print now · Shift per job.
Job history (v2.7.0)	Full manager: **filter** by status (incl. one-click “Not printed”), **search** title/printer/#id, load 50/100/300/1000, paging, **tick rows** (check / uncheck all) with a live “N selected” counter, **Reprint selected**, **Reprint all non-printed**, **Clear ALL history** (finished jobs only — waiting jobs are never touched), per-row Retry / Reprint.
Audit trail	Latest events: who printed what from where, reprint reasons, token errors.
Live debug & test tool + Hub settings	**Send test print & watch live** (job → printer verdict), **Release ALL held jobs**, **Full self-test (A-to-Z)**, SumatraPDF path setting. **v2.7.0:** both panels now sit properly INSIDE the page (they used to hang below it, unstyled and full-width).

6. Printing everything

6.1 PDFs (the lab-report flow)

- **Simple mode** (default ON): any PDF opened in the browser prints silently and the tab closes.
- **Rules**: URL pattern + optional title filter + optional printer → auto-prints the moment the URL opens (glob or regex).
- **From your app's code**: `printPdfLink(url, 'Lab report', {fallback:false})` — the page API prints via the hub and never opens a dialog. Full API: extension `content/api.js`, docs at `/docs`.

6.2 Web pages

Right-click → AdvPrint Pro → Print page · `Alt+P` · popup button · `window.advprint.printPage()`. Rendered with headless Chrome/Edge at print quality. `window.print()` and Ctrl+P are intercepted (toggle in Settings) and routed silently.

6.3 Receipts / raw text (ESC/POS)

Send `kind:"raw"` or `|center|…|` markup to `/print`. Codepage selectable (cp437/cp850/cp866/cp1251/cp1252). Bangla and other complex scripts are automatically rendered through headless Chrome so they shape correctly on thermal rolls (`thermal_text_engine`).

6.4 Selections & screenshots

Right-click a selection to print exactly it; the popup can print the visible page as an image — useful for POS systems and stubborn apps.

6.5 Save as PDF instead of paper

Choose printer `File (save as PDF)` or send `"printer":"__file__"`. Files land in `~/Downloads/AdvPrint` or the folder set in Hub settings (`file_dir`).

7. LAN / IP — share one printer with every PC

1. On the **printer PC**: extension Options → LAN / PCs → **Enable LAN sharing** (or dashboard → LAN card). Restart the hub.
2. Firewall: allow inbound TCP 9235 (Windows: `netsh advfirewall firewall add rule name="AdvPrint Pro" dir=in action=allow protocol=TCP localport=9235` — run once as admin).
3. On each **client PC**: install the extension, set **Hub URL** to `http://<printer-PC-ip>:9235`, press **Detect token** on the printer PC (or paste the token) → Connect.
4. Every PC now prints to the shared printers; jobs can be **shifted** between PCs (single job or the whole waiting queue) when a PC goes down.

The token is required from every LAN PC — no token, no printing (the hub rate-limits and locks out foreign IPs after too many bad tokens, while your own PC (127.0.0.1) is **never** locked out by default).

8. Queue, History & Audit trail

- **Queue mode ON**: jobs wait in FIFO order, Pause/Resume works, one job prints at a time.
- **Queue mode OFF (instant)**: each job prints immediately; if the printer is not active the job is **auto-queued** (`instant_auto_queue`) and prints when the printer is back.
- **Printer-status hold** (`hold_offline`): jobs whose printer reports offline/paper-out are held with a reason and auto-print on recovery — or hit **Release ALL held jobs** / per-job **Print now** to force.
- **Retries**: failed jobs retry with exponential backoff (`max_retries` × `retry_base_seconds`).
- **Duplicates**: identical jobs inside `dedupe_minutes` are blocked (double-click protection).
- **History — v2.7.0 manager**: last 50/100/300/1000 jobs in the hub DB (SQLite), per-job file for reprints (`keep_files`); in the dashboard AND in Options → History: **filter** by status, **search**, **tick** rows, **Reprint selected**, **Reprint all non-printed**, **Clear ALL history** (finished jobs only — queued/held jobs survive). Export CSV from the dashboard. API: `POST /jobs/bulk-reprint` (`{"ids":[…]}` or `{"scope":"not_printed"}`), `POST /history/clear` .
- **Audit trail**: who printed what, from which site/IP, reprint reasons, auth failures — CSV export too; disable with `audit_enabled:false` .

9. Settings reference — every switch in one table

Change from: extension Options → Hub settings (live) · dashboard · `~/.advprint/config.json` · environment variables (`ADVPRINT_*` ; legacy `PRINTHUB_*` still honored).

Key	Default	Meaning
token	random per install	API token — rotate via /token/rotate. Old default PrintHub-2024 kept on legacy upgrades.
token_header_only	false (true on fresh installs)	Strict mode: POST calls must send the X-AdvPrint-Token header.
allowed_origins	[] (all)	Restrict which websites may print, e.g. ["https://www.htncr.org"].
max_auth_failures / lockout_minutes / lockout_exempt_loopback	10 / 5 / true	Lockout policy for wrong tokens; loopback exempt by default (owner can never lock himself out).
printer	"	Force a printer; empty = OS default; "__file__" = save as PDF.
printer_groups	{}	{"Front Desk": ["A","B"]} — fallback printers in order.
profiles	{}	{"Invoice": {"paper":"a4","duplex":true}} — named option sets; send "profile":"Invoice".
queue_mode (extension)	true	FIFO queue vs instant print.
max_retries / retry_base_seconds	2 / 30	Exponential retry of failed jobs.
dedupe_minutes	5	Duplicate block window.
hold_offline	true	Hold jobs while the printer reports offline/out of paper.
instant_auto_queue	true	Instant mode falls back to queue when no printer is active.
share_lan	false	Listen on 0.0.0.0 so other PCs can print (token always required).
peer_hubs / peer_forwarding	[] / false	Other PCs' hubs for queue shifting / auto-forwarding.
thermal_text_engine / escpos_codepage	auto / cp437	Raw text rendering (Bangla → browser engine) and ESC/POS codepage.
audit_enabled / reprint_watermark / reprint_reason_required	true / false / false	Audit trail + compliance switches for reprints.
file_dir / keep_files	~/Downloads/AdvPrint / true	Save-as-PDF destination; keep files for reprint.
sumatra_path	"	Manual SumatraPDF.exe path (empty = auto-find).
debug	false	Verbose [DEBUG] logs + client error reports in /debug.
log_max_mb / log_backups	5 / 3	hub.log rotation.

10. Debug & the Live Test tool

Four layers, from one-click to full control:

1. **Full self-test (A-to-Z)** — v2.6: extension Debug tab button · dashboard “Full self-test” · `AdvPrint Tool` menu [11] · CLI `python advprint.py livetest` (add `print` to also queue a real test page). Checks: Python, data folder writable, token, SumatraPDF/lp, browser, port, queue DB mechanics, printers, default printer — each line `[OK]` or `[!!]` with the fix. Exit code 0 = production-ready.
2. **Test print & watch live** — dashboard button queues a real page and follows the job to the printer with a verdict (REAL PRINTER HIT / SAVED / FAILED + reason). Tray icon → Print test page. CLI `testprint`.
3. **Debug mode** — verbose logs + browser error reports; the Debug tab and `/debug` answer "WHY is nothing printing?" with concrete problems + fixes.
4. **Bundles for support** — `/diagnostics`, `/health`, `/debug` JSON, audit CSV, history CSV, "Copy report" buttons. Token is redacted in diagnostics.

```
$ python advprint.py livetest
  [OK] Python >= 3.8                       3.11.9
  [OK] Data folder + history.db writable   /home/you/.advprint
  [OK] API token configured                private random token
  [OK] CUPS lp command                     /usr/bin/lp
  [OK] Chrome/Edge (page printing)         /usr/bin/chromium
  [OK] Port 9235 available                 OK
  [OK] Queue DB enqueue/read/update        job #42
  [OK] Printers detected                   HP_LaserJet, XP-80C
  [OK] Default printer                     HP_LaserJet

  ALL CHECKS PASSED - AdvPrint Pro is ready for production.
```

11. Enterprise edition — silent install, policy, service

11.1 Silent / scripted Windows install

```
AdvPrint-Pro-Setup-2.7.0.exe /S          # silent, per-user, no UAC
# enterprise rollout via Intune/GPO/SCCM: run per user, no elevation needed
```

11.2 Managed policy (force settings on all PCs)

Deploy `extension/managed.json` content as Chrome policy `HKLM\SOFTWARE\Policies\Google\Chrome\ExtensionInstallForcelist` + managed storage, or simply push the keys. Forceable settings: `hubUrl`, `token`, `defaultPrinter`, `enabled`, `queueMode`, `retryOfflin e`, `lang`, `hookPrint`, `autoPrintAllPdfs`, `failFallback`, `closeAfterPrint`, `rules` (rules fully override local ones — central lockdown). Managed values win over user settings.

11.3 Windows service mode (headless)

```
python advprint.py service-install     # UAC → registers services.msc: "AdvPrint Pro Service"
python advprint.py service-status / service-remove
```

Service mode = no tray, runs before logon, perfect for a print-server PC. The normal Startup app stays the default on desktops; if both run, the second exits gracefully (port guard).

11.4 Lockdown checklist

- `allowed_origins` → only your web app may print.
- `token_header_only:true` + rotated token on every PC.
- `share_lan:false` everywhere except the print-server PC; that one keeps `lockout_exempt_loopback` and strict lockouts for foreign IPs.
- `audit_enabled:true` + `reprint_reason_required:true` for compliance; audit CSV ships out nightly.
- LAN firewall rule only where needed (§7).

12. HTTP API quick reference

Auth: header `X-AdvPrint-Token` (legacy `X-PrintHub-Token` accepted), or `?token=` on GET navigation. Full docs: `docs/API.md` and the hub's own `/docs` page.

```
# status
curl http://127.0.0.1:9235/health
curl -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/printers
# print a PDF from bytes (silent)
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -H "Content-Type: application/pdf" \
     --data-binary @report.pdf "http://127.0.0.1:9235/print?title=Report"
# print a page with options
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -H "Content-Type: application/json" \
  -d '{"url":"https://intranet/roster","paper":"a4","duplex":"duplex","copies":2}' \
  http://127.0.0.1:9235/print
# queue control / tokens / settings
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/queue/pause
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/token/rotate
curl -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/config
# live test over HTTP (same check as the CLI)
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/livetest
```

Page-side API (in any web page once the extension is installed): `printPdfLink(url, title, opts)` · `window.advprint.printPdf/printPage/printers/status/history`.

13. Troubleshooting — every known problem and its fix

Symptom	Cause → Fix
Setup says *cannot access / Access is denied*	Incomplete/damaged download (pre-2.6 bug) or Mark-of-the-Web → re-download the 2.7.0 zip (setup ≈ 9.6 MB), right-click zip → Properties → Unblock, or run `install-windows.bat` which unblocks automatically.
Dashboard shows “needs the current token”	Stale bookmark → press the green **Fix it now** button on that very page (hub PC), or tray → Open dashboard, or extension → Detect token.
Extension shows 401 / token rejected	Options → Settings → **Detect token** (or paste the token from `advprint token`). After a rotation always re-detect.
Nothing prints, no error	Debug tab → Run full self-test → fix the [!!] line (usually SumatraPDF missing or no default printer). Then send a test print & watch live.
Jobs stuck “queued”	Queue paused → Resume; printer held → **Print now** or Release ALL held jobs; check printer status in the dashboard Printers card.
Jobs stuck “held”	That is the printer-status hold doing its job — fix the printer or **Print now**; disable hold in Hub settings (`hold_offline`).
Bangla/receipt text is garbage	Hub settings → thermal engine `auto` (browser render) + correct codepage; browser printing needs Chrome/Edge installed.
LAN PC cannot connect	`share_lan` ON + hub restarted + firewall TCP 9235 + correct Hub URL + token; check the LAN/PCs tab shows the PC's URL.
“python” in Task Manager	You started with system python — the installer's `AdvPrintPro.exe` (renamed pythonw) shows “AdvPrintPro”; re-run setup or the Startup shortcut.
Auto-run missing	Task Manager → Startup apps → “AdvPrint Pro” enable; or tray menu → Start at Windows logon.
Old PrintHub Pro traces after upgrade	Setup 2.6 removes the old Startup app/task/service/registry keys automatically; leftovers: uninstall the old version once, or delete `%LOCALAPPDATA%\PrintHubPro`.
Hub shows OFFLINE / start “did not answer”	Open the desktop tool → **[13] Diagnose & fix**: it unblocks the folder (Mark-of-the-Web), frees port 9235 from zombies, then starts the hub in plain view so the real error shows as text. Ends with the hub.log tail + a copy/paste start command with the full path.
Tool prints the diagnosis but hub.log is empty	The hub died before logging (AV quarantined the fresh exe) → add the AdvPrint folder to the antivirus exclusions, run [13] again (it unblocks), then [1] Start.
“python advprint.py serve” says *can't open file*	You are in the wrong folder — that command needs the file's real location. Use the tool's copy/paste command (menu [13], step [5/5]) or just menu [1].
Dashboard tables show “-” and buttons do nothing	Old-build JS bug (one broken quote killed the whole dashboard script) — **fixed in 2.6.4**; update, hard-refresh the dashboard (Ctrl+F5).
Tray icon missing	Fixed in 2.6.4 (embedded-runtime isolated mode). Update + start the hub; the icon sits next to the clock — may hide under the `^` arrow. If the hub is not running there is no tray — start with [1] first.
Hub will not start (port busy)	Another instance runs — check the tray icon; the hub exits gracefully with a clear message instead of crashing; [13] can free the port for you.
Screen “glitch” / black console flashes every few seconds	Old build (pre-2.6.6): the hub polled the service state with `sc query` without the windowless flag — Windows flashed a console each time. **Fixed in 2.6.6** — every background child now runs windowless; update.
`save-as-PDF failed: [Errno 13] Permission denied`	A real-time antivirus / indexer briefly locked the fresh PDF. **Fixed in 2.6.7** — the hub now writes atomically and retry-reads ~4 s. If it still repeats: add `C:\Users\<you>\.advprint` to your AV exclusions, then **Retry** the job from History.
`print backend exit 1:` with no message (File (PDF))	**Fixed in 2.6.7** — “File (PDF)” is now a virtual printer: the PDF is **saved** to `Downloads\AdvPrint` with status “saved” (also with Queue Mode OFF). The backend's stdout is captured too, so real printer errors now show a readable reason.
Windows service “AdvPrint” stuck in *start pending*	A stale state left by an older service process — the hub actually runs from the Startup app / tray. Fix: `services.msc` → AdvPrint → Restart (or reboot), or run `service-remove` — keep only ONE of service / Startup app active. The Debug tab lists this automatically.

`printer 'X...X not found'` — the name shows TWICE in a row (e.g. `CanonCanon`)	A stored printer name was typed/pasted twice. **Self-healed in 2.6.8** — the hub collapses the doubled name to the real printer, prints, and corrects the history (audit: `printer.corrected`). Old dead jobs: just press **Retry**. Still check Settings → Printer for the stale entry.
Every page prints TWICE — two sheets, and the log shows two `[OK] job #N` lines	Queue Mode OFF race in v2.6.8 and earlier: the FIFO worker and the direct thread both grabbed the same job. **Fixed in 2.6.9** — the job row is created in its final state, one job = one sheet. Not a printer or driver problem (the printer was innocent). Upgrade the hub to 2.6.9+; nothing to change on the printer.
Right-click "Print this page" fails with *Cannot access contents of url chrome-extension://…*	The menu was used on a browser-internal page (chrome://, settings, extension page). **Fixed in 2.6.8** — the menu now only appears on printable pages (http/https/file) and shows a friendly notice otherwise.
HP Laser 103/107/108 shows *offline*	That is the printer device itself (USB/network/sleep), not a hub error — check cable/power, set it online in Windows, then **Retry** the held jobs.

14. Upgrade, uninstall, files & folders

14.1 Upgrade

Run the new setup over the old one (same per-user location, settings/history kept) — or replace the portable folder. The extension updates by Remove + Load unpacked (or GitHub update check → new zip). Data folders migrate: `~/.printhub` → `~/.advprint` automatically.

14.2 Uninstall

Windows: Start Menu → Uninstall AdvPrint Pro (or Apps & features). Service: removed automatically. History/settings in `~/.advprint` are kept (delete manually if unwanted). Linux: `sudo apt remove advprint-pro` + `uninstall-unix.sh`.

14.3 Where everything lives

What	Path
Program (Windows)	`%LOCALAPPDATA%\AdvPrintPro` (hub, runtime, install scripts, extension, docs)
Data (config, history.db, hub.log, files)	`%USERPROFILE%\.advprint` (migrated from `.printhub`)
Auto-run	Startup app “AdvPrint Pro” (Task Manager) · optional service “AdvPrint” (services.msc)
Program (Linux deb)	`/opt/advprint` + `/usr/bin/advprint*` · data `~/.advprint`
Logs	`~/.advprint/hub.log` (rotating)

You are never locked out: the hub never locks out your own PC (127.0.0.1) by default, the tray always opens the dashboard with the right token, and the error page itself offers the one-click fix. That is the v2.6 promise: **auto security, zero manual Security-tab surgery**.

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